

MTH203: Sequences and series

Lecture 08

1. AN EXAMPLE

We look at an example of a sequence that does not converge.

Example 1. Let $(x_n)_{n \in \mathbb{N}}$ be the sequence of real numbers, defined by $x_n = (-1)^n$. The first few terms of this sequence can be written as follows:

$$-1, 1, -1, 1, -1, 1, \dots$$

In other words, $x_n = -1$ if n is odd and $x_n = 1$ if n is even.

We claim that $(x_n)_n$ diverges. Indeed, suppose this sequence converges to some point $x \in \mathbb{R}$. We will show that there exists some ϵ for which there does not exist any natural number N such that $x_n \in V_\epsilon(x)$ for every $n > N$.

Indeed, let us take $\epsilon = 1$. Suppose that there exists some N such that for any $n > N$, we have $|x - x_n| < 1$. Take n to be any odd number greater than N . Then $x_n = -1$ and so $|x - (-1)| < 1$. Since $n + 1$ is even, we have $x_{n+1} = 1$. Since $n + 1 > n > N$, we must have $|x - 1| < 1$. Then, the triangle inequality implies

$$1 + 1 > |x - (-1)| + |x - 1| = |(-1) - x| + |x - 1| \geq |(-1) - 1| = |-2| = 2.$$

This is a contradiction. Thus, we see that the sequence $(x_n)_n$ diverges.

It is very important to understand this example clearly. Note that 1 occurs infinitely many times in this sequence. Thus, $|1 - x_n| = 0$ for infinitely many values of n . However, for any $N > 0$, if we take m to be an *odd* integer greater than N , then $|1 - x_m| = 2$. Thus, no “tail” of this sequence can end up in a sufficiently small neighbourhood of 1. For a similar reason, -1 also cannot be the limit of this sequence even though it appears in the sequence infinitely many times.

2. UNIQUENESS OF LIMITS

The idea in the example above can be used to prove the following theorem:

Theorem 2. Let S be a metric space. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in S and let a, b be elements of S such that $(x_n)_{n \in \mathbb{N}}$ converges to both a and b . Then $a = b$.

Proof. Suppose $a \neq b$. Then $|a - b| > 0$. Let ϵ be a positive real number such that $\epsilon < |a - b|/2$. (For example, we could take $\epsilon = |a - b|/4$.)

Since $(x_n)_{n \in \mathbb{N}}$ converges to a , there exists an integer N_1 such that for any $n > N_1$, we have $|a - x_n| < \epsilon$. Similarly, since $(x_n)_{n \in \mathbb{N}}$ converges to b , there exists an integer N_2 such that $n > N_2$, we have $|b - x_n| < \epsilon$. Let $N = \max\{N_1, N_2\}$. Then, if $n > N$, we have $|a - x_n| < \epsilon$ as well as $|b - x_n| < \epsilon$. Adding these two inequalities, we get

$$|a - x_n| + |b - x_n| < \epsilon + \epsilon < |a - b|/2 + |a - b|/2 = |a - b|.$$

This contradicts the triangle inequality for a, b and x_n . Thus, $a = b$. □

Thus, if $(x_n)_{n \in \mathbb{N}}$ is known to be a convergent sequence, the notation $\lim_{n \rightarrow \infty} x_n$ unambiguously refers to a single element of the metric space.

3. BOUNDED SEQUENCES

A sequence $(x_n)_{n \in \mathbb{N}}$ in a metric space S is said to be *bounded* if it is contained in an open ball of the form $V_R(x)$ for some $x \in S$ and some positive real number R . Recall that

$$V_R(x) = \{y \in S \mid d(x, y) < R\}.$$

Thus, a sequence is bounded if its terms do not stray too far from some fixed point in the metric space.

Lemma 3. A sequence $(x_n)_{n \in \mathbb{N}}$ in $\mathbb{R}$ is bounded (with respect to the standard metric) if and only if there exists a positive real number M such that $|x_n| \leq M$ for all n .

Proof. If there exists a positive real number M such that $|x_n| \leq M$ for all n , then $d(0, x_n) \leq M$ for all n (where d denotes the standard metric on $\mathbb{R}$). Thus, the sequence is bounded.

Conversely, suppose $(x_n)_{n \in \mathbb{N}}$ is a bounded sequence. Thus, there exists an $x \in \mathbb{R}$ and some positive real number R such that $|x - x_n| < R$ for all n . Then,

$$|x_n| = |x_n - 0| \leq |x_n - x| + |x - 0| < R + |x|$$

for all n . Thus, the condition in the lemma is satisfied with $M = R + |x|$. This completes the proof. $\square$

Theorem 4. Every convergent sequence in a metric space is bounded.

Proof. Let S be a metric space and let $(x_n)_{n \in \mathbb{N}}$ be a sequence such that $\lim x_n = x$. Then, there exists an integer N such that $n > N$ implies $d(x_n, x) < 1$.

For every integer i such that $1 \leq i \leq N$, let $d_i = d(x, x_i)$. Let $M = 1 + \max\{d_1, \dots, d_N, 1\}$. For every positive integer n such that $n \leq N$, we have $d(x, x_n) = d_n < d_n + 1$ and so $d(x, x_n) < M$. On the other hand, if $n > N$, we have $d(x, x_n) < 1 < M$. Thus, we have $d(x, x_n) < M$ for all positive integers M as required. $\square$

The converse of this theorem is *not true* in general.

Example 5. Let $(x_n)_{n \in \mathbb{N}}$ be the sequence of real numbers defined by $x_n = (-1)^n$. This sequence is clearly bounded. However, as we saw in the last lecture, this sequence does not converge.

We will see a partial converse to this theorem in the next section.

4. MONOTONE SEQUENCES

We set up some terminology regarding sequences of real numbers. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence of real numbers.

- (1) We say that $(x_n)_{n \in \mathbb{N}}$ is *non-decreasing* if $x_{n+1} \geq x_n$ for all n . (Sometimes a non-decreasing sequence is called an *increasing* sequence. However, this adjective can sometimes be misinterpreted to mean that we have a strict inequality $x_{n+1} > x_n$. So, many people prefer to use the word “non-decreasing”.)
- (2) Similarly, we say that the sequence is *non-increasing* if $x_{n+1} \leq x_n$.
- (3) A sequence that is non-decreasing or non-increasing is said to be *monotone*.
- (4) A sequence is *bounded above* if there exists a real number M such that $x_n \leq M$ for all n .
- (5) A sequence is *bounded below* if there exists a real number M such that $x_n \geq M$ for all n .

Here is a very easy exercise:

Exercise 6. Prove that a sequence of real numbers is bounded (with respect to the standard metric) if and only if it is bounded above as well as bounded below.

We have the following *partial* converse to Theorem 4.

Theorem 7. A bounded, monotone sequence is convergent.

Proof. Let $(x_n)_{n \in \mathbb{N}}$ be a bounded monotone sequence. Let us assume that this is a non-decreasing sequence. (The case of the non-increasing sequence is similar.)

The sequence $(x_n)_{n \in \mathbb{N}}$ is bounded above. Thus the set $S := \{x_1, x_2, \dots\}$ has an upper bound, and hence it has a supremum x . We claim that $\lim x_n = x$.

Indeed, since x is the supremum of S , given any $\epsilon > 0$, there exists an integer N such that $x_N > x - \epsilon$. Then, if $n > N$, we have $x_n > x_N$ and so $x - x_n < x - x_N < \epsilon$. On the other hand, as $x_n < x$, we also have $0 < x - x_n$. Thus, $|x - x_n| < \epsilon$ for any $n > N$. This completes the proof. $\square$

5. READING SUGGESTIONS

This lecture corresponds to part of Section 2.3 of the textbook ([1]).

REFERENCES

- [1] Stephen Abbott, *Understanding Analysis*, Undergraduate Texts in Mathematics, Springer, 2015. [3](#)